MILITARY SPECIFICATION

TEST SET, COMPUTER, DEVIATED PURSUIT

AN/APM-207

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be capable of testing the operation and accuracy of, and facilitate maintenance of Computer, Deviated Pursuit CP-742/APQ. The unit shall contain the necessary circuits to supply primary power, functional controls, simulated data inputs and readout devices for the computer circuits.

1.2 Classification - Test Set, Computer, Deviated Pursuit AN/APM-207 shall be of one type and shall consist of the following items:

Items	Type Designation	Appl. Para.
Test Set, Computer, Deviated Pursuit	TS-2039/APM-207	3.4.1
Cable Assembly Power, Electrical	W1 of AN/APM-207	3.4.1.10
Cable Assembly, Special Purpose, Electrical	W2 of AN/APM-207	3.4.1.11
Cable Assembly, Special Purpose, Electrical	W3 of AN/APM-207	3.4.1.12
Cable Assembly, Special Purpose, Electrical	W4 of AN/APM-207	3.4.1.13
Lead, Electrical	W5 of AN/APM-207	3.4.1.14
Case, Test Set	CY-4214/APM-207	3.4.1.5

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-R-23373	Radar Set AN/APQ-94, Military Specification for
MIL-R-21642	Radar Set AN/APQ-83, Military Specification for
MIL-C-3432	Cable and Wire, Electrical (Power and Control; Flexible and Extra Flexible, 300 and 600 Volts)
MIL-E-4682	Electron Tubes and Transistors, Choice and Application of

FSC 4920

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MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-D-18300	Design Data Requirements for Contracts Covering Airborne Avionic Equipment
MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates, for Aeronautical Electronic and Associated Equipment
MIL-T-21200	Test Equipment, for use with Electronic and Fire Control Systems, General Specification for
MIL-W-16878	Wire, Electrical, Insulated, High Temperature
MIL-T-23336	Technical Directives (Letter Type): Preparation of

Federal

TT-E-527	Enamel, Alkyd, Lusterless

2.2 Availability of Documents

(1) When requesting specifications, standards, drawings and publications refer to both title and number.

(2) Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120, Attn: Code 105).

2.3 Precedence of Documents - When the requirements of the contract, this specification or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials the following shall govern:

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(1) Parts and materials requirements shall conform to Specification MIL-T-21200.

(2) Nonrepairable subassemblies, as outlined in Specification MIL-T-21200, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. A mean-time-to failure of more than 5000 hours should be the goal of each.

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

(4) The contractor shall report to the Bureau all instances wherein selective assembly of electronic components, which comply with military specifications, are required to achieve assembly performance requirements.

3.1.1 Nonstandard Parts and Material Approval - For approval for the use of nonstenderd parts end materials see 6.2(3)(e).

3.1.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied, and the complements reported as indicated in paragraph 6.2(3)(b).

3.2 Design and Construction

3.2.1 General - The equipment shall conform to all the applicable requirements of Specification MIL-T-21200 for Class 3 equipment for design, construction and workmanship, unless otherwise specified herein. The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 42 pounds.

3.2.2 Operating Life

3.2.2.1 Operational Stability - The equipment shall operate with optimum performance for 24 hours, continuously or intermittently, without the need to readjust any control.

3.2.2.2 Reliable Operating Life - The equipment shall have a reliable operating life of at least 200 hours without removal for bench servicing. Parts requiring servicing or replacement at the end of this interval, to renew this service life, shall be specified by the manufacturer.

3.2.2.3 Total Operating Life - The equipment shall have a minimum total operating life of 2000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.3 Cabling and Connectors

3.2.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-T-21200.

3.2.4 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-T-21200.

3.2.5 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

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3.2.6 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-T-21200. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.7 Nomenclature and Nameplates - For nomenclature assignment and nameplate approval for equipment identification see 6.2(3)(c).

3.2.8 Standard Conditions - The following conditions shall be used as a basis to establish normal performance requirements and for making laboratory bench tests on the equipment.

Temperature	Room ambient ($30^{o}C \pm 10^{o}C$)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input Voltage	115 ±1.0 vac and 27.0 ±0.5 vdc

3.2.9 Sand, Dust, Fungus and Explosion - Sand, dust, fungus and explosion requirements of Specification MIL-T-21200 shall not be applicable to this equipment.

3.2.10 Service Conditions - The equipment shall operate satisfactorily under any of the environmental and mechanical service conditions or reasonable combination of these conditions as specified in Specification MIL-T-21200 for Class 3 equipment, except as modified herein.

3.2.10.1 Vibration - In accordance with Specification MIL-T-21200, paragraph 3.2.14.4 with these exceptions:

Frequency Range	Double Amplitude
5 to 15	0.06 inch
15 to 25	0.04 inch
25 to 33	0.02 inch

A test at 0.02 inch Double Amplitude and the Rate of Frequency varying between 33 to 55 cycles per second is to be run for exploratory determination of points of high mechanical transmissability. This test is to be nondestructive.

3.2.11 Input Power Requirements - The equipment shall meet all applicable requirements of Specification MIL-T-21200, Class 3, and shall give specified performance from the following power sources with characteristics as defined herein. The power required shall not exceed the specified amounts.

(1) AC Power (Three Phase)

115 ±10% vac

Operating Voltage Limits	103.5 to 126.5 vac
Operating Frequency Limits	380 to 420 cps
Power	125 va max

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(2) DC Power 27 +2.0 vdc

Operating Voltage Limits	25 to 29 vdc
Current	1 amp max

3.2.11.1 Overload Protection - Overload protection for the equipment shall be provided in the equipment. All parts and circuits of the equipment, which are likely to carry an overload due to any failures or poor adjustment, shall have suitable protective devices or shall be proportioned to withstand such overload with permanent damage to the equipment. The use of fuses and other devices shall be held to a minimum.

3.2.11.2 Undervoltage Protection - The equipment shall not be damaged by operation at voltages down to 98 vac and/or 21 vdc and shall automatically resume normal operation when the voltage returns to within the specified operating limits.

3.2.12 Data Requirements - No data are required by this specification, or by applicable documents referenced in Section 2, unless specified in the contract or order (see paragraph 6.2).

3.3 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.3.1 Operation - The Test Set, Computer, Deviated Pursuit shall be capable of providing input and control signals for performing computer lead determination, slaving and navigation circuitry checks.

3.4 Detail Requirements

3.4.1 Test Set, Computer, Deviated Pursuit AN/APM-207

3.4.1.1 Functions - The test set shall be capable of testing the operation and accuracy of Computer, Deviated Pursuit CP-742/APQ as herein specified.

3.4.1.2 Form Factor - The Test Set, Computer, Deviated Pursuit AN/APM-207 shall not exceed 19-5/8 inches wide, 14-1/2 inches high, 13-3/4 inches deep and shall not exceed 42 pounds, excluding cables.

3.4.1.3 Case, Test Set CY-4214/APM-207 - The combination case shall conform to Specification MIL-T-21200, except it shall be watertight only with the cover in place.

3.4.1.4 Finish - The case shall be finished with gray enamel conforming to Specification MIL-T-21200. The front panel shall be finished with black synthetic lusterless enamel in accordance with Specification TT-E-527.

3.4.1.5 Cover - The combination case shall be supplied with a detachable cover in accordance with Specification MIL-T-21200. The cover shall provide storage for instruction books and the accessories specified in paragraph 3.4.1.9. The cover shall not exceed 19-5/8 inches wide, 5 inches high, and 13-3/4 inches deep.

3.4.1.6 Controls and Indicators - The following controls and indicators, labeled as specified, shall be provided on the front panel of the Test Set, Computer, Deviated Pursuit AN/APM-207. The tolerances specified are for conditions of paragraph 3.2.8 only. For conditions of paragraph 3.2.10 increase all tolerances by additional 0.5 percent.

(1) POWER - A toggle switch with positions labeled "ON," "OFF" and "RESET" shall be provided on the front panel to connect and disconnect the incoming primary power.

(2) ROLL - A toggle switch with positions labeled "60°" and "0°" shall be provided to permit selection of one of two simulated vertical gyro roll signals to the computer under test.

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(3) MISSILE ID-A three-position toggle switch with positions labeled "LEFT", "NONE", and "RIGHT" shall be provided to connect +28 vdc to the Offside Relay and Lead Enable Relay in the computer, thus decreasing the lead angle when an offside missile is selected.

(4) RANGE - A toggle switch with positions labeled "LOCK ON" and "UNLOCKED" shall be provided to connect +28 vdc to the Range Locked On Relay, thus readying the computer for lead computation.

(5) MISSILE - Toggle switch with positions labeled "ON" and "OFF shall be provided to connect +28 vdc to the Sarah ID Relay in the computer, thus readying the computer for Sarah lead computation.

(6) MODE - A toggle switch with positions labeled "LEAD" and "NO LEAD" shall be provided to connect +28 vdc to the No Lead Relay in the computer, thus readying the computer for lead computation. A no-lead condition is indicated when this relay is deenergized.

(7) ωa Switch - A three-position toggle switch with positions labeled "+ωa", "OFF" and "-ωa" shall be provided for selection of plus, minus or no azimuth rate signals to the computer under test.

(8) ωa Control - A ten-turn precision potentiometer with an indicating dial shall be provided to furnish an azimuth rate signal between zero and ±1 vdc. This control shall operate in conjunction with the ωa switch. The signal voltage shall be equal to the dial reading in millivolts and between 0.0 to 60 millivolts shall be capable of being adjusted to an accuracy of 0.002 vdc. Adjustment accuracy shall be 10% up to 0.5 vdc and 20% up to 1.0 vdc.

(9) λe Switch - A four-position rotary switch with positions labeled "+ λe", "- λe", "RANGE" and "AZ" shall be provided for selection of simulated plus or minus radar antenna position deviation signals or radar range.

(10) λe Control - A ten-turn precision potentiometer with an indicating dial shall be provided to furnish antenna elevation signals between zero and ±75 vdc or range signals between zero and +75 vdc. This control shall operate in conjunction with the λe switch. The signal voltage shall be equal to the dial reading divided by 13-1/3. This is equivalent to $\frac{\text{D. R. x 3}}{40}$ and shall be capable of being adjusted to an accuracy of 1.5% or 0.2 vdc whichever is greater.

(11) λa Switch - A four-position rotary switch with positions labeled "+λa", "-λa", "R MAX" and "R RATE" shall be provided for selection of plus or minus simulated radar antenna position azimuth signals, missile R max or R rate signals.

(12) λa Control - A ten-turn precision potentiometer with an indicating dial shall be provided to furnish antenna azimuth signals between zero and ±75 vdc, and R max or R rate signals between zero and +75 vdc. This control shall operate in conjunction with the λa switch. The signal voltage shall be equal to the dial reading divided by 13-1/3. This is equivalent to $\frac{\text{D. R. x 3}}{40}$ and shall be capable of being adjusted to an accuracy of 1.5% or 0.2 vdc whichever is greater, for all cases except R max. In setting R max the capability of accuracy will be 4% or 0.2 vdc whichever is greater.

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(13) VOLTMETER - A 3-1/2 inch, sealed and ruggedized, panel voltmeter shall be provided. It shall have a 50 - 0 - 50 microamp linear movement and shall function as a zero center scale voltmeter or as a null indicating meter. When used as a voltmeter the meter shall have an initial accuracy of ±2% of full scale.

(14) VOLTMETER POLARITY SWITCH - A three-position rotary switch with positions labeled "+VOLTS", "VOLTMETER" and "-VOLTS" shall be provided to permit selection of voltmeter or plus or minus null meter functions for positive or negative signals to the VOLTMETER.

(15) METER FUNCTION SWITCH - A 24-position rotary switch shall be provided to permit selection of the input signal voltages to the VOLTMETER. The switch shall have 24 positions labeled numerically 1 through 24. Rotation shall be continuous with no stops to limit travel. Position 12 shall provide for front panel input to the VOLTMETER through a test point.

(16) METER RANGE SWITCH - A four position rotary switch shall be provided to permit selection of the full scale voltage to be indicated on the VOLTMETER. The four positions shall be labeled:

"50 VOLTS"	".5 VOLTS"
"5 VOLTS"	".05 VOLTS"

(17) REFERENCE ADJUST - A ten-turn precision potentiometer with an indicating dial shall be provided to permit adjustment of the reference voltage used to null the VOLTMETER. The range of adjustment shall be between zero and ±75 vdc. Polarity of the reference voltage shall be determined by the position of the VOLTMETER POLARITY SWITCH. The reference voltage shall be equal to the dial reading divided by 13-1/3. This is equivalent to $\frac{D.R. \times 3}{40}$ and shall be capable of being adjusted to an accuracy of 1.0% or 0.075 vdc whichever is greater.

(18) POWER Lamp - A panel lamp shall be provided to indicate when primary power is applied to the test set.

(19) COINCIDENCE Lamp - A panel lamp shall be provided to indicate coincidence during the navigation constant circuit check.

3.4.1.7 Computer Power Supply - When supplied with primary power as specified in paragraph 3.2.11, the Test Set, Computer, Deviated Pursuit shall provide the following primary power voltages to the computer under test:

(1) 115 vac ±10%, 380 to 420 cps, 3 phase, 4 wire

(2) +150 vdc ±4% 16 ma

(3) -250 vdc ±3% 4 ma

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3.4.1.8 Reference Power Supplies - When supplied with primary power as specified in paragraph 3.2.9, the Test Set, Computer, Deviated Pursuit shall provide the following reference voltages:

(1) +75 vdc ±1% 50 ma

(2) -75 vdc ±1% 50 ma

These power supplies shall provide the reference and signal voltages adjusted by means of the ω a control, λ e control, λ a control and REFERENCE ADJUST control.

3.4.1.9 Cables - The equipment shall operate with the following cables, which shall be supplied as part of the equipment.

3.4.1.10 Cable Assembly, Power, Electrical W1 - A 6-foot power cable to provide primary power to the Test Set, Computer, Deviated Pursuit. One end of the cable shall be terminated with a connector identified as P1304 for connecting to the test set. The branched ends of the cable shall be terminated with the following connectors:

(1) A connector identified as P1305 shall be provided for connecting to an external +27 vdc power source.

(2) A connector identified as P1306 shall be provided for connecting to an external 115 vac, 400 cps power source.

3.4.1.11 Cable Assembly, Special Purpose, Electrical W2 - A 3-foot test cable to provide connection between the Test Set, Computer, Deviated Pursuit and the computer under test. The connectors shall be identified as P1301 and P4001.

3.4.1.12 Cable Assembly, Special Purpose, Electrical W3 - A 3-foot test cable to provide connection between the Test Set, Computer, Deviated Pursuit and the computer under test. The connectors shall be identified as P1302 and P4002.

3.4.1.13 Cable Assembly, Special Purpose, Electrical W4 - A 3-foot test cable to provide connection between the Test Set, Computer, Deviated Pursuit and the computer under test. The connectors shall be identified as P1303 and P4003.

3.4.1.14 Lead, Electrical W5 - A 4-foot 2-inch test lead to provide connection between the front panel test point and external voltage test points.

3.4.1.15 Connectors - Four multi-pin connectors shall be provided on the front panel for primary power and test connections. The input power connector shall be identified as J1304. The test connectors shall be identified as J1301, J1302 and J1303.

3.4.1.16 Fuses - The following fuses shall be provided for circuit protection in the event of electrical overload. A spare fuse shall be provided for each value and located as listed.

System	Number	Value	Location	Spare Location
115 vac Input	1	1 amp	Front Panel	Cover of Case
+28 vdc Input	1	1 amp	Front Panel	Cover of Case

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System	Number	Value	Location	Spare Location
75 Volt Supply	2	3/16 amp	Internal	Rear of Chassis
250 Volt Supply	1	.010 amp	Front Panel	Cover of Case
150 Volt Supply	1	1/32 amp	Front Panel	Cover of Case

3.5 Preproduction Sample - Unless otherwise specified in the contract or order, a preproduction sample of the Test Set, Lead-Launch Computer shall be manufactured such that it is representative of the production lot in all respects. The sample will be tested as specified in section 4 herein and is for the purpose of determining that the contractor, prior to his starting production, is capable of yielding items that comply with the technical requirements of the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

(1) Preproduction Tests

(2) Production Sample Tests

(3) Acceptance Test

(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration test shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. Contractors not having adequate facilities to conduct all required tests shall obtain the service of a commercial testing laboratory satisfactory to the procuring activity.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requested by the procuring activity, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

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4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipment under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment. It may not be resubmitted as the production sample model.

4.3 Production Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent to a designated government laboratory for tests. The equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality, and that it meets all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Upon failure of the production sample to pass any of the tests delivery of equipment under the contract shall cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when required by the procuring activity.

4.3.3 Reconditioning of Production Test Sample - After completion of the production sample tests the equipment shall be returned to the contractor for any necessary rework. The equipment shall then be resubmitted for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring agency. Acceptance or approval of material during the course of manufacture shall not be

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construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Reliability Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Reliability Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3 g's.

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting

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plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in two directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Sampling tests shall be conducted on equipment which have successfully passed the individual tests. The following equipments shall be selected by the Government Inspector for sampling tests:

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 10	1
Next 50	1
Next 75	1
Next 100	1
1 for each additional 200 or fraction thereof	

When a defective unit from a group occurs, no items from this group, or later produced groups, shall be accepted until the extent and cause of failure has been determined and appropriately corrected. In addition, when a failure occurs, shift to one sample out of the first ten (10) and proceed as indicated when inspection is resumed.

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, production and special tests.

(3) Manufacturing reliability test specified in paragraph 4.4.1.3, except that the test duration shall be 30 hours.

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4.4.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early production equipment.

(2) On an early equipment after an engineering change.

(3) Whenever failure reports or other information indicate additional equipment should be tested. (This will be determined by the procuring activity.)

4.4.3.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable.

4.4.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in paragraph 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following, except that when a production sample equipment is requested and submitted to the procuring activity for test then the first equipment specified below need not be selected or subjected to this test. (Equipments which have successfully passed the sampling test or special tests may be selected for life test.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Life Test
First 25	1
Next 175	1
Next 300	1
1 for each additional 500 or fraction thereof	1

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4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (0 - 5000 feet)
Humidity	Room ambient
AC Voltage	115 (+2.5 -6.5) volts (at lowest applicable frequency)
DC Voltage	27 ±2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor may resubmit the equipment for acceptance.

4.6 Test Procedures - The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts, such procedures may be used when approved by the procuring activity.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555. For level A packaging, the method of preservation shall conform to Specification MIL-P-116, Method II.

6. NOTES

6.1 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.0 ±0.5 VDC and 115 ±1.0 VAC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

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6.2 Ordering Data - Procurement documents should specify the following:

(1) Title, number and date of this specification.

(2) Selection of applicable level of packing and packaging required (see 5.1).

(3) Data required for operational use and for maintenance and reprocurement purposes. Specific data to be considered include:

(a) Approval for use of nonstandard part and material shall be obtained as outlined in Specification MIL-E-5400.

(b) Tube complement report shall be in accordance with Specification MIL-E-4682 (see 3.1.2).

(c) Nomenclature assignment and nameplate approval for equipment indentification shall be in accordance with Specification MIL-N-18307 (see 3.2.7).

(d) Data obtained by the contractor in conducting Contractor's Demonstration Tests shall be submitted to the procuring activity for review and approval prior to shipping the equipment to the specified destination for service approval tests. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired.

(e) The contractor shall submit the accessory material and design, and test data specified in Specification MIL-E-5400. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements for service approval.

(f) The contractor shall prepare test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable.

(g) A record shall be kept of all failures. (See 4.4.1.3)

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(h) When test procedures previously approved by production and sampling tests are not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

1. Temperature - Altitude Tests
2. Vibration Test (60 minutes cycling in each of 3 planes)
3. Humidity Tests

(i) Submit to the procuring activity for approval the proposed corrective action intended to reduce the possibility of the same failure (s) occurring in future tests.

(j) The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all transistor failures. This record shall indicate the following:

1. Part type number
2. Circuit reference symbol number
3. Part function
4. Name of manufacturer
5. Nature of failure
6. Number of hours part operated prior to failure.

(k) The procedures used for conducting preproduction, acceptance, and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval.

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(1) When a defective unit is found, acceptance of all units after the last group of accepted units, shall be withheld until the contractor shall:

1. Determine the cause of the defect.

2. Take the necessary action to correct or eliminate the defect from all units on hand and, if so directed by the procuring activity, from those units previously shipped.

3. Submit to the procuring activity for approval the proposed corrective action to prevent recurrence of the defect.

4. Return to one sample out of the first 25 and proceed as indicated when inspection is resumed.

6.3 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ±0.5 vdc and 115±1.0 vac. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration.

6.5 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-T-81173(WP) TEST SET, COMPUTER, DEVIATED PURSUIT AN/APM-207

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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Washington, D. C. 20360

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